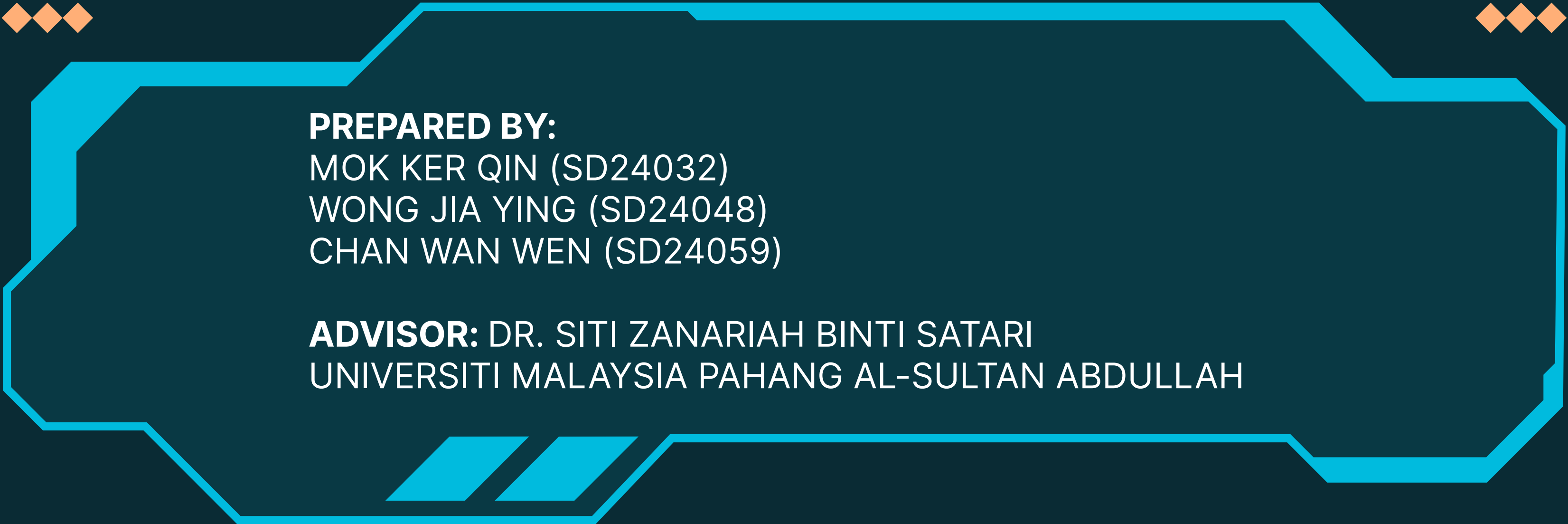




THE TALENT TRAP : **PREDICTING GRADUATE** **UNDEREMPLOYMENT RISK IN MALAYSIA**



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PROBLEM STATEMENT

Malaysian graduates face a growing talent crisis driven by structural inequality rather than wages alone. This study identifies three key factors:

Expectation gap

- Many graduates, especially Diploma holders, earn far less than what average salary figures suggest, leading to disappointment and a sense of being misled.

Rural penalty

- Graduates working in rural areas often receive lower pay, pushing them to move to cities and leaving local communities short of talent.

Unequal Opportunities

- While Degree holders see improving outcomes, Diploma holders continue to face limited progression and higher risks of underemployment.

Malaysia Labour Market Insight Series. (2025). The Gaji Cukup Makan Economy: When Higher Education Becomes an Economic Risk. <https://euera.org/wp-content/uploads/The-Gaji-Cukup-Makan-Economy.pdf>

SIGNIFICANT OF STUDY

Sustainable Development Goals (SDG):

- **SDG 4 Quality Education:** Highlights mismatch between education level and employment outcomes, especially for Diploma graduates.
- **SDG 8 Decent Work & Economic Growth:** Identifies underemployment risks to promote fair wages and decent job opportunities.
- **SDG 10 Reduced Inequalities:** Reveals wage gaps across education levels and urban-rural locations.

Environmental, Social, and Governance (ESG):

- **Social (S):** Addresses inequality, rural penalty, and underemployment risks among graduates.
- **Governance (G):** Uses transparent, explainable tree-based models (Decision Tree & Random Forest) for data-driven policy decisions.

United Nations. (2025). Sustainable Development Goal 4: Quality Education. Retrieved from Sustainability Magazine. https://sustainabilitymag.com/articles/the-united-nations-sustainable-development-goals?utm_source

United Nations. (2025). Sustainable Development Goal 8: Decent Work and Economic Growth. Retrieved from <https://www.un.org/sustainabledevelopment/sdgbookclub-8-archive/> un.org

United Nations. (2025). Sustainable Development Goal 10: Reduced Inequalities. Retrieved from <https://sdgs.un.org/goals/goal10>

GRADUATE STATISTICS 2020-2024 (DOSM)

Jadual 10g: Purata Gaji & Upah Bulanan Siswazah Bekerja mengikut Kumpulan Umur, Malaysia, 2020 - 2024					
Table 10g: Mean Monthly Salaries & Wages of Employed Graduates by Age Group, Malaysia, 2020 - 2024					
Strata	2020 ^r	2021 ^r	2022 ^r	2023	2024
Jumlah	4,466	4,542	4,747	4,933	5,330
Total					
Bandar	4,697	4,767	4,906	5,209	5,428
Urban					
Luar Bandar	3,664	3,758	3,846	4,019	4,283
Rural					
Ijazah	5,492	5,519	5,607	5,979	6,247
Degree					
Bandar	5,656	5,680	5,698	6,069	6,335
Urban					
Luar Bandar	4,657	4,732	4,722	4,834	5,172
Rural					
Diploma	3,172	3,343	3,533	3,769	4,080
Diploma					
Bandar	3,308	3,480	3,615	3,784	4,160
Urban					
Luar Bandar	2,606	2,694	2,924	3,024	3,339
Rural					

Jadual 10h: Penengah Gaji & Upah Bulanan Siswazah Bekerja mengikut Sijil Tertinggi Diperoleh dan Strata, Malaysia, 2020 - 2024					
Table 10h: Median Monthly Salaries & Wages of Employed Graduates by the Highest Certificate Obtained and Strata, Malaysia, 2020 - 2024					
Strata	2019	2020	2021	2022 ^p	
Jumlah	3,711	3,956	4,265	4,409	4,521
Total					
Bandar	3,928	4,140	4,403	4,637	4,706
Urban					
Luar Bandar	3,026	3,164	3,391	3,426	3,503
Rural					
Ijazah	4,661	4,825	5,024	5,146	5,724
Degree					
Bandar	4,763	4,938	5,075	5,340	5,724
Urban					
Luar Bandar	4,226	4,368	4,207	4,352	4,587
Rural					
Diploma	2,796	2,823	3,072	3,121	3,390
Diploma					
Bandar	2,942	2,969	3,149	3,291	3,454
Urban					
Luar Bandar	2,185	2,201	2,403	2,417	2,723
Rural					

Dosm. (n.d.). Department of Statistics Malaysia. <http://statistics.gov.my/portal-main/release-content/graduates-statistics-2024>

TRAINING DATASET

Based on official statistics

- The data is synthesized using the median salary values from the DOSM Graduate Statistics 2024 Report.
- This ensures the data is centered on what a typical graduate earns, avoiding the skew caused by extremely high earners.

Realistic salary simulation

- The data is synthesized using a Stratified Truncated Normal Distribution.
- Groups graduates by Education and Location to capture realistic income inequality.
- This method centers salaries on DOSM median values while strictly enforcing an RM 1,500 floor to comply with the 2024 Minimum Wage Order.

Defining ‘High Risk’

- Established the risk threshold using the Belanjawanku Expenditure Guide (2022/2023) (Urban: RM 3,500 / Rural: RM 3,000), adding a RM 900 buffer for savings and emergencies.

Adding real-world unpredictability

- Inserted 15% random label noise
- This flips the outcome for 15% of graduates to simulate hidden factors (like debt or family obligations) and prevents the AI from just memorizing the rules.

Sample of Synthesized Training Data

Row No.	Student_ID	Education	Location	Salary	Risk_Status
1	1001	Diploma	Urban	3293	High Risk
2	1002	Degree	Urban	4914	Stable

Norma, D., Nik, M., Ainoon, N., & Osman, N. (2022). BELANJAWANKU EXPENDITURE GUIDE -A REFERENCE BUDGET FOR MALAYSIAN. https://rmke12.ekonomi.gov.my/ksp/storage/event/124_24_puan_nik_noor_ainoon_belanjawanku_referece_budget_for_malaysian.pdf

RESEARCH QUESTIONS

1. What insights on trend variations can be derived from visualizing graduate underemployment risk data in Malaysia?
2. How accurate are predictive models for graduate underemployment risk in Malaysia when developed using the Decision Tree and Random Forest algorithms?
3. Which tree-based machine learning model, specifically Decision Tree and Random Forest model, that provides the highest accuracy in predicting graduate underemployment risks in Malaysia, and how can it be optimized for better results?

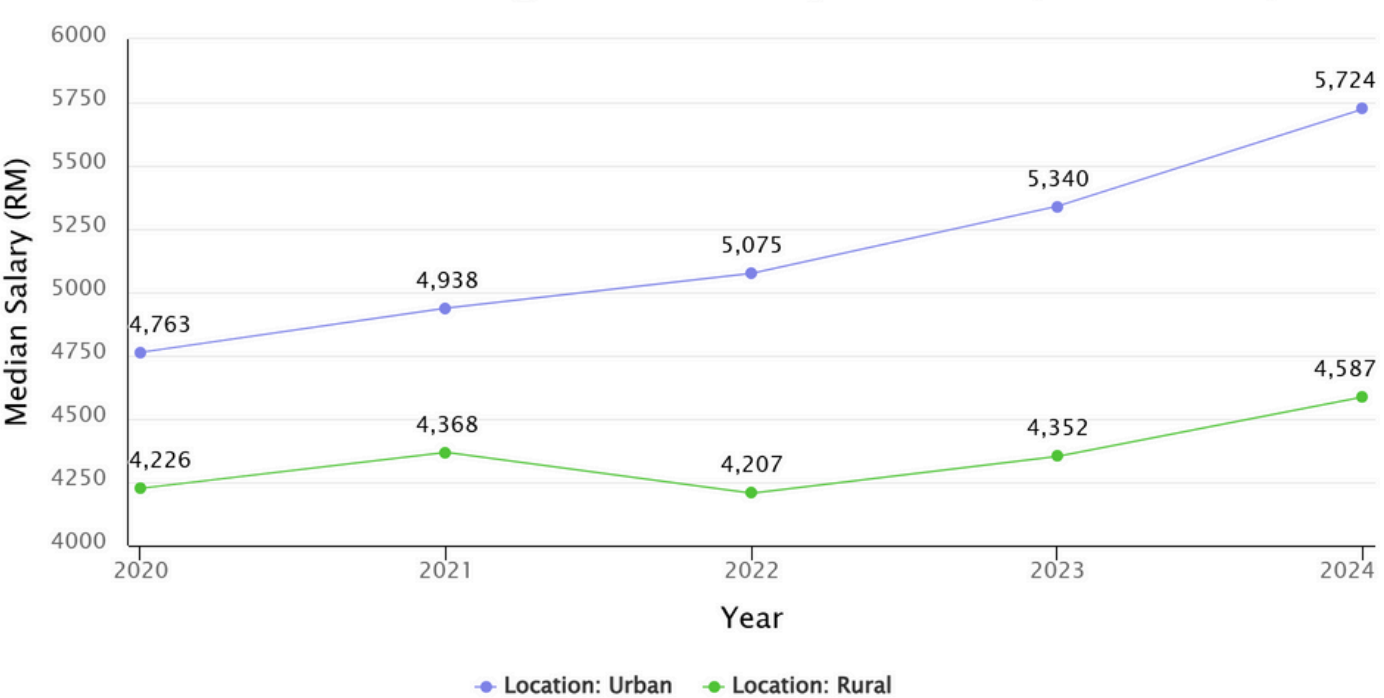
RESEARCH OBJECTIVES

1. To visualize and analyze trend variations and demographic patterns of graduate underemployment risk in Malaysia.
2. To develop tree-based predictive models for graduate underemployment risk using Decision Tree and Random Forest algorithms in Altair AI Studio.
3. To predict the graduate underemployment using the best tree-based model with higher accuracy.

Objective 1: To visualize and analyze trend variations and demographic patterns of graduate underemployment risk in Malaysia.

MEDIAN SALARIES: DEGREE VS DIPLOMA

Median Salaries of Degree Holders by Location (2020–2024)



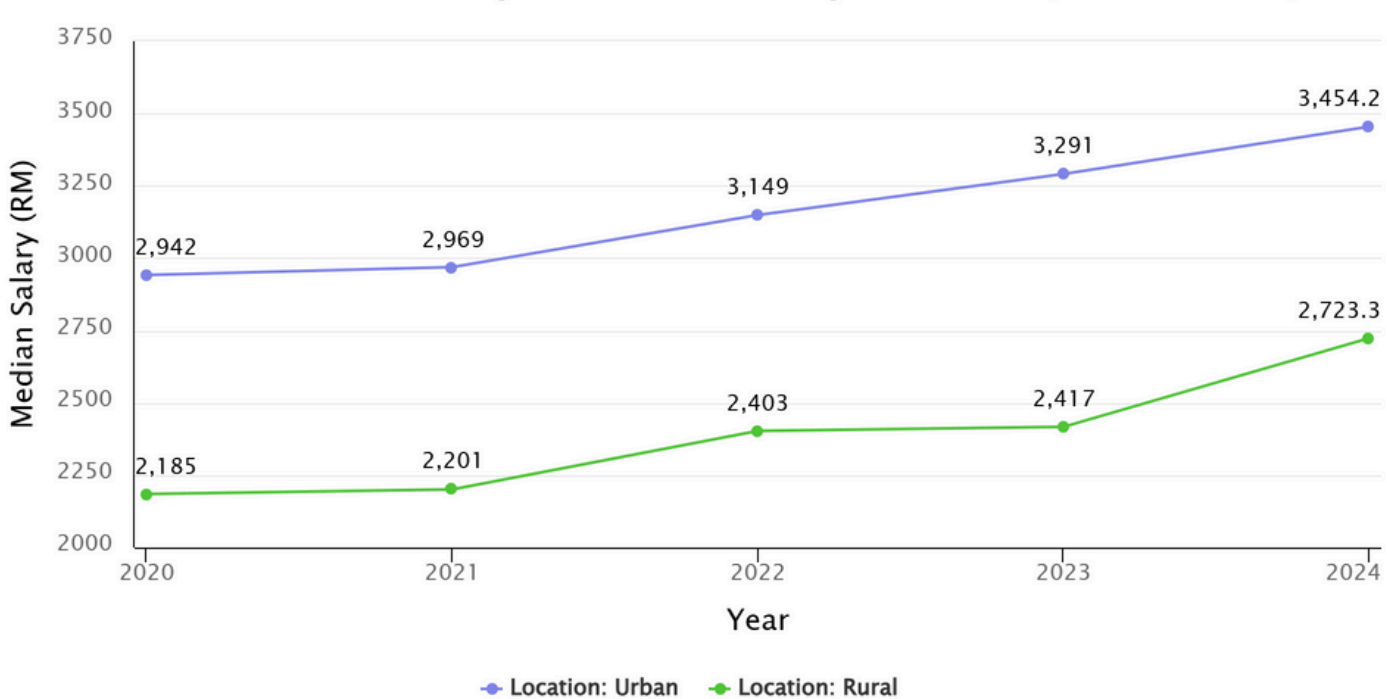
Urban–rural gap:

- Wage gap widened from RM537 (2020) to RM1,137 (2024).
- Urban graduates consistently earn more, with the disparity increasing over time.
- Median salaries for degree holders are consistently higher in urban areas than in rural areas, indicating a persistent location-based advantage.
- The dip in 2022 for rural degree holders is likely due to post-pandemic recovery lag. Rural job markets recovered more slowly after COVID-19, with fewer high-paying roles reopening compared to urban areas.

Urban–rural gap:

- Wage gap widened from RM757 (2020) to a peak of RM874 (2023) before narrowing to RM731 (2024).
- Urban graduates consistently earn more than rural graduates, but the wage gap fluctuates over time.
- The persistent gap suggests that location limits access to higher-paying jobs, reinforcing ongoing inequality for rural graduates.
- Rural diploma graduates continue to face greater income inequality and structural disadvantage compared to their urban peers.

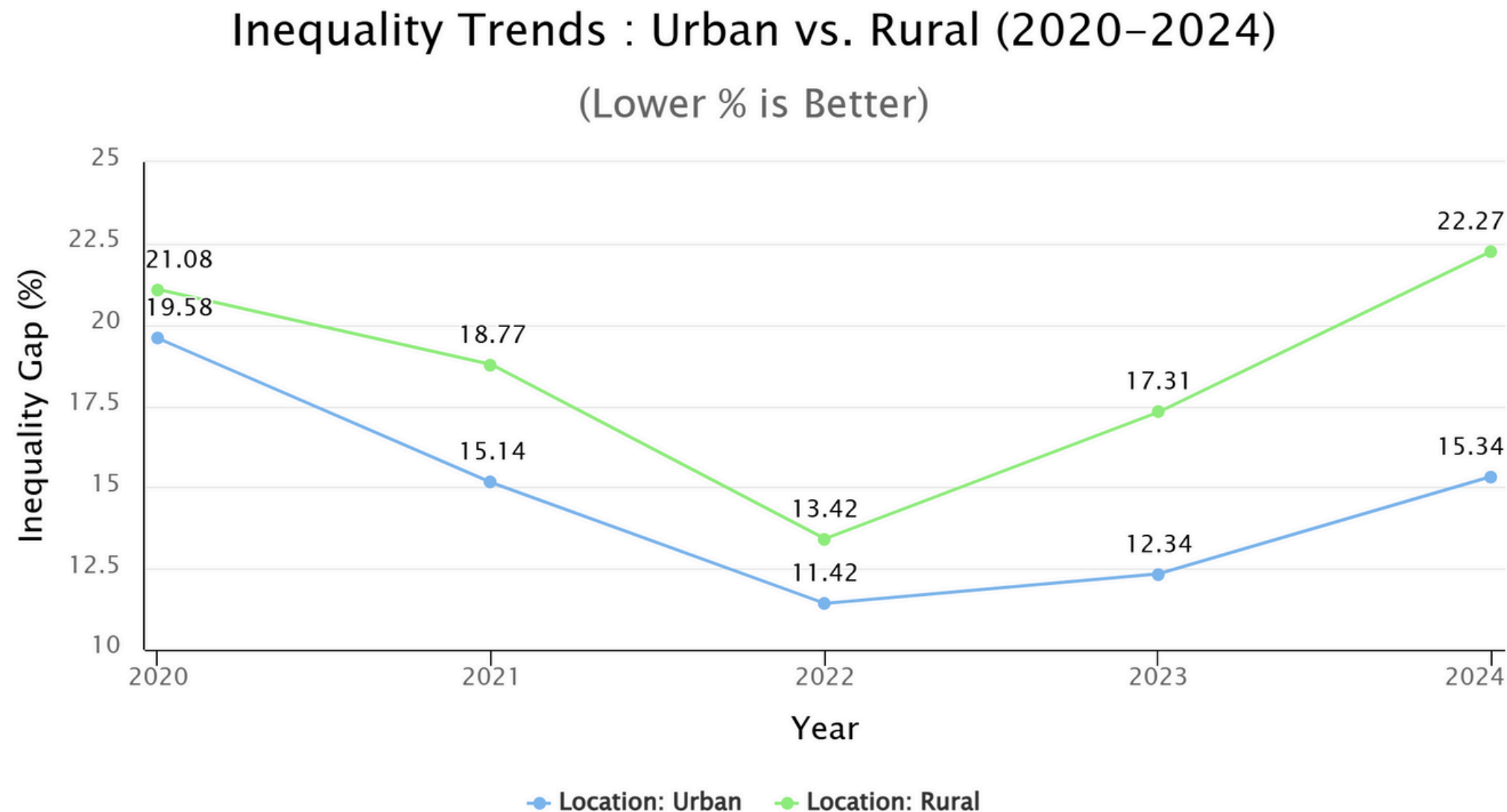
Median Salaries of Diploma Holders by Location (2020–2024)



Degree holders earn significantly more than diploma holders in both urban and rural areas. The urban–rural salary gap is more severe for diploma holders. Location matters for all graduates, but diploma holders are more vulnerable to income inequality than degree holders. This shows that higher education provides partial protection against location-based income inequality.

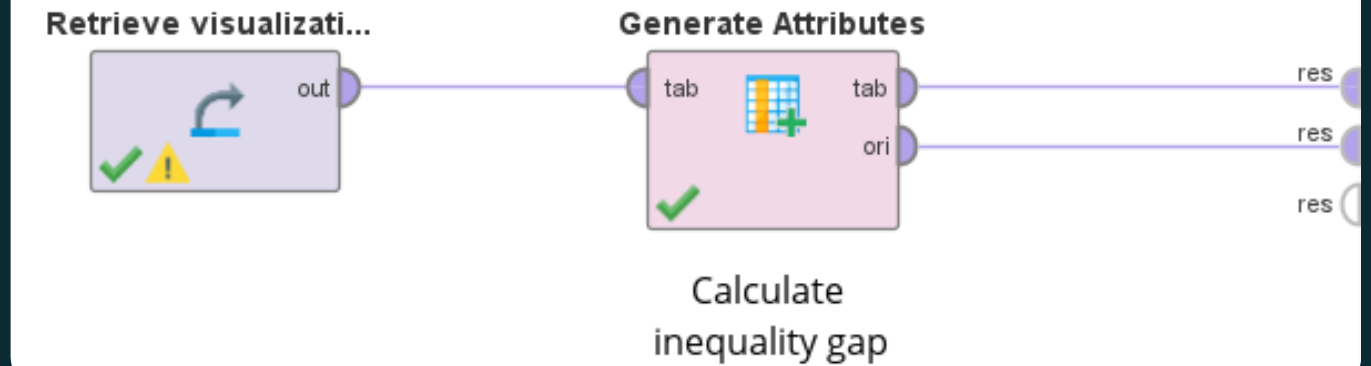
Objective 1: To visualize and analyze trend variations and demographic patterns of graduate underemployment risk in Malaysia.

INEQUALITY TRENDS: URBAN VS RURAL



$$\text{Inequality Gap} = \frac{\text{Mean} - \text{Median}}{\text{Median}} \times 100$$

This metric calculates how much the average salary is artificially inflated by high-income outliers compared to what a typical graduate actually earns.



Overall trend:

- **Inequality declines** from 2020 to 2022 for both locations, indicating improved income balance.
- After 2022, inequality rises again, especially in rural areas.

Abnormal pattern:

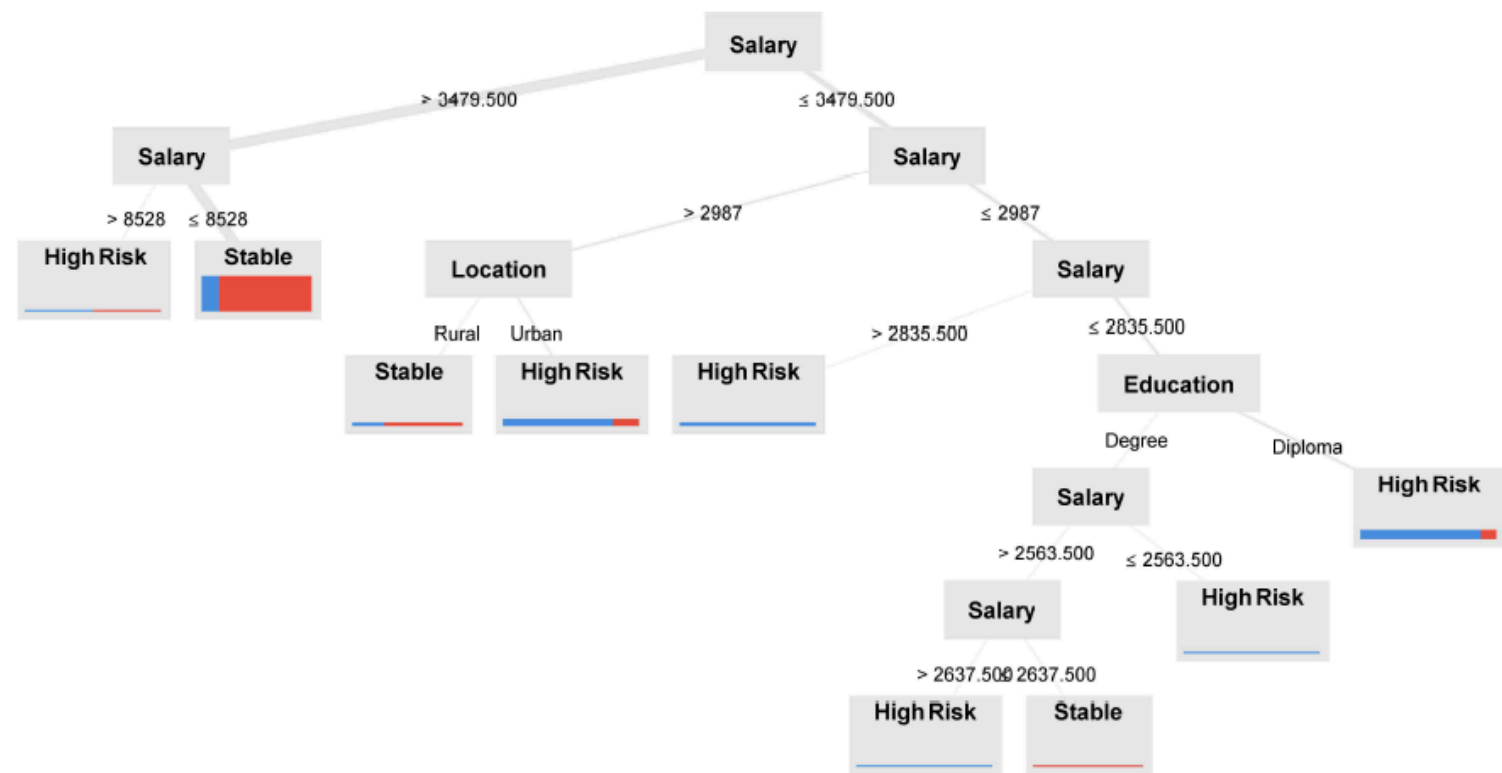
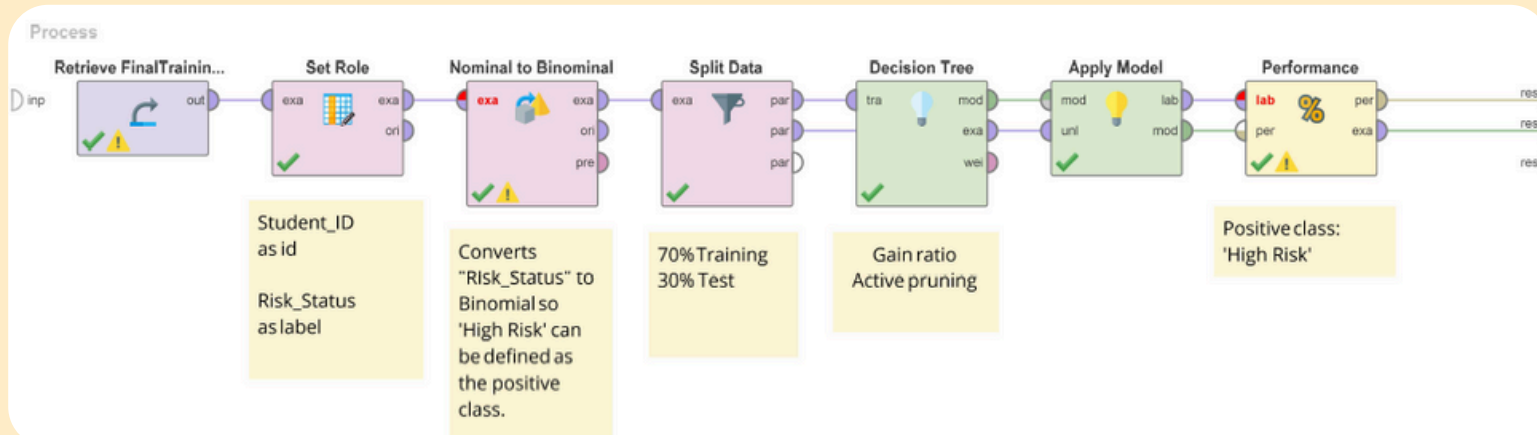
- Inequality declines sharply in 2022 for both groups, then rises again afterward.

Insight:

- Rural graduates face greater income inequality, highlighting ongoing disadvantages compared to urban graduates.

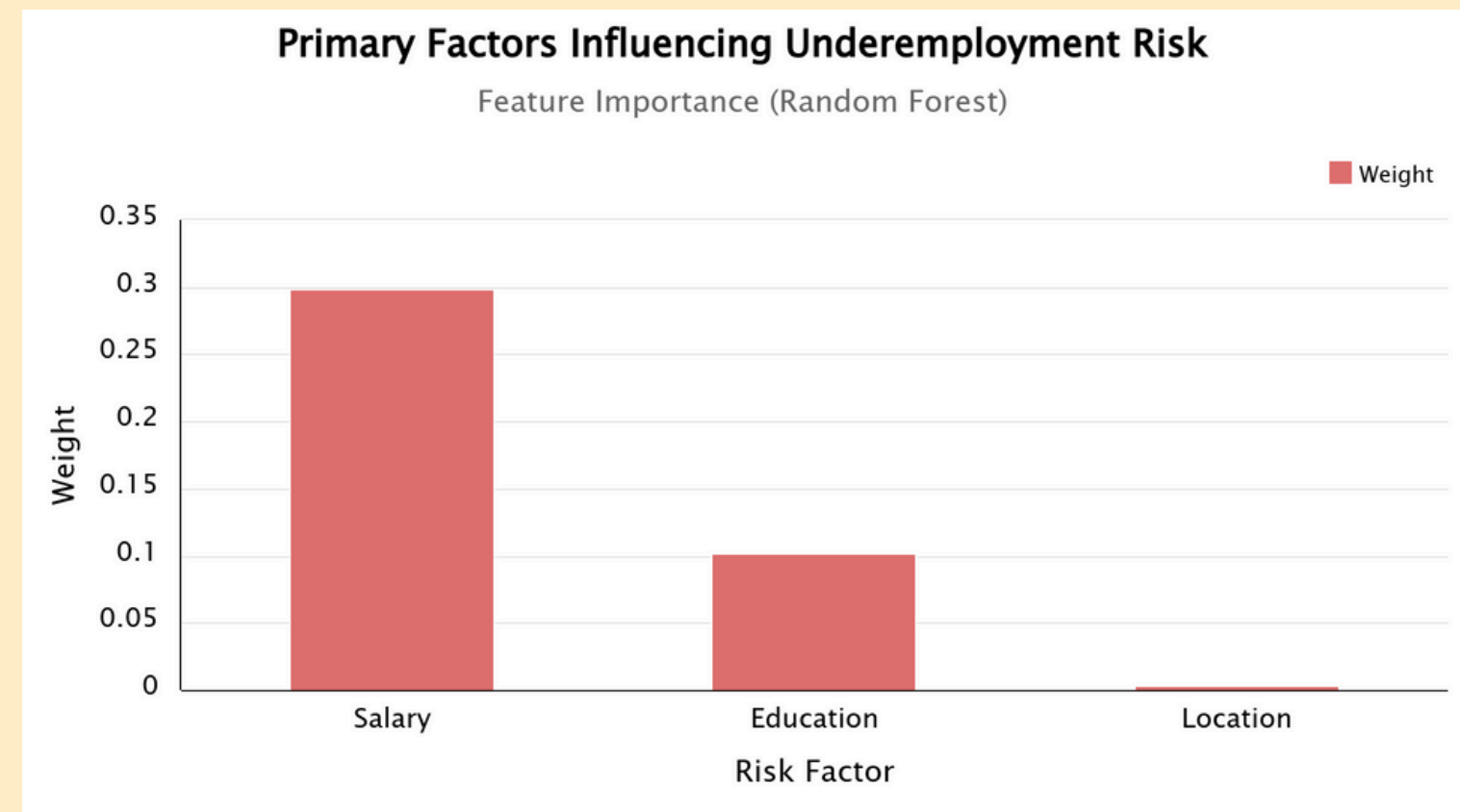
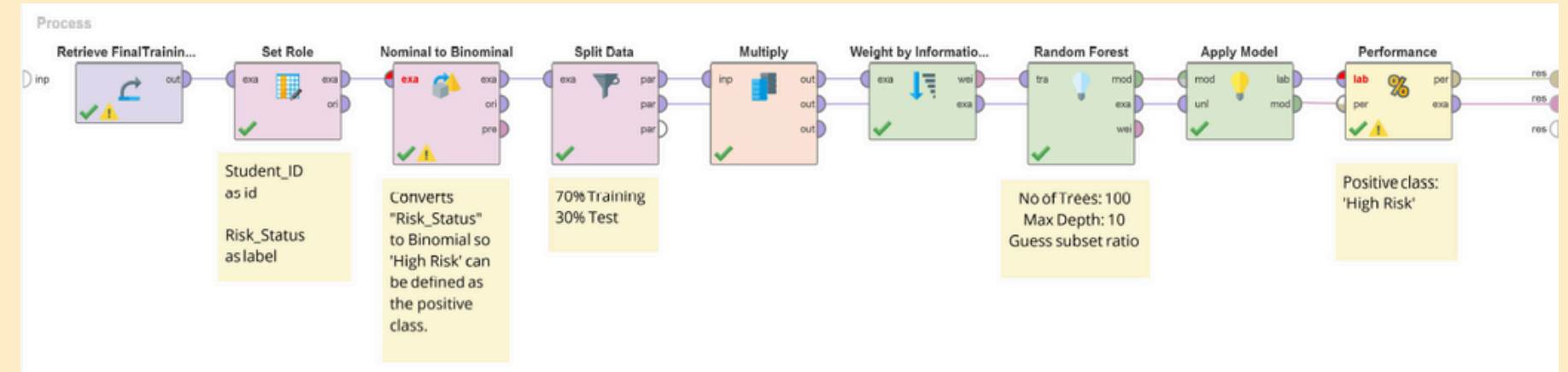
Objective 2: To develop tree-based predictive models for graduate underemployment risk using Decision Tree and Random Forest algorithms in Altair AI Studio.

DECISION TREE



The Decision Tree identified that **RM2,835.50** is the exact salary cut-off needed to be financially 'safe'. Anyone earning below that is very likely to be in financial trouble, regardless of their location and education level.

RANDOM FOREST



The Random Forest identified that **Salary (0.30)** is the primary factor influencing the risk of underemployment. **Location (0.003)** has a near-zero global impact, debunking the idea of "rural means poor".

COMPARISON BETWEEN THE DECISION TREE AND THE RANDOM FOREST MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.85	0.86	0.71	0.78
Random Forest	0.83	0.84	0.70	0.76

- **Decision Tree model outperforms the Random Forest model:**
 - With higher accuracy, precision, recall, and F1-score.
 - Indicates it has a better ability to accurately classify graduates into “High Risk” and “Stable” categories.
- **Decision Tree performs better because:**
 - The dataset is simple and has fewer variables.
 - Effectively captures the direct relationship between Salary and Risk Status.
- **Random Forest performance may be affected by:**
 - Computational noise from the ensemble method.
 - Unnecessary complexity for this dataset.

Considering these metrics, the best-performing model is the **Decision Tree model**.

SIMULATOR

Decision Tree - Simulator

Education: ⓘ

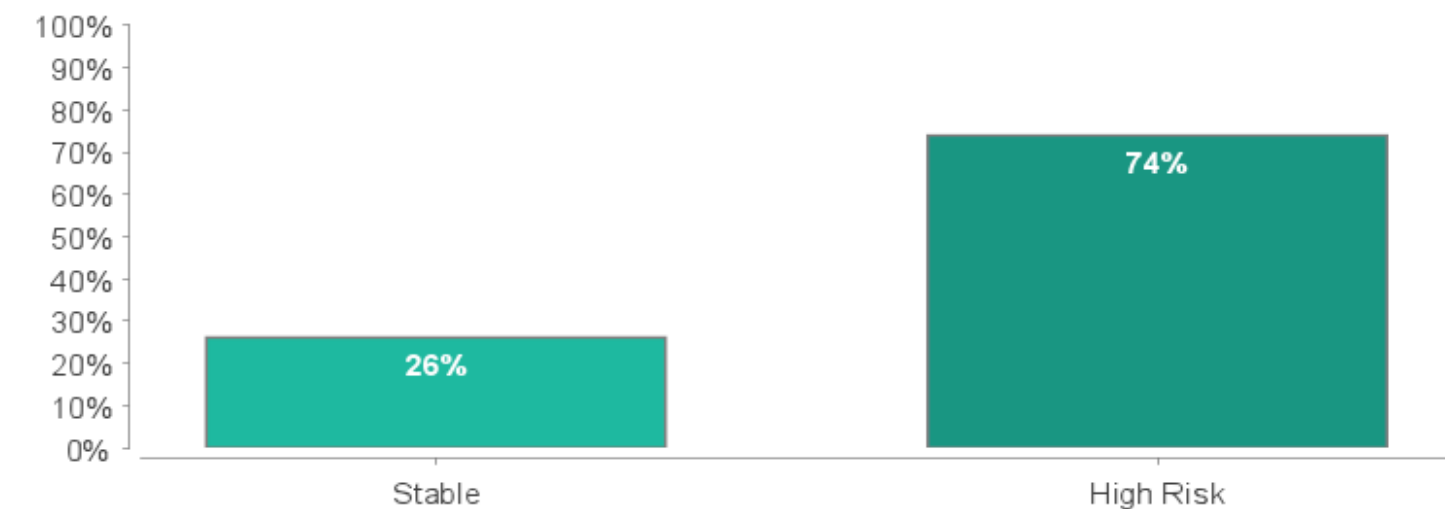
Location: ⓘ

Salary: ⓘ

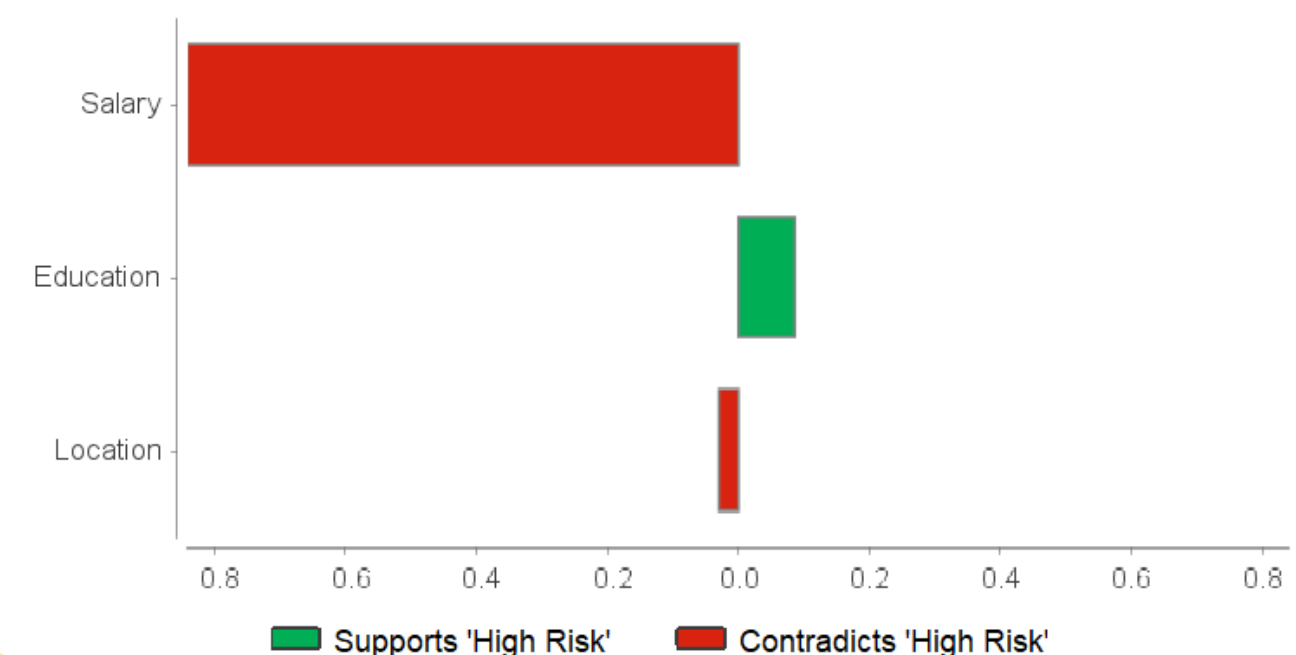
The simulator reveals that an **Urban Degree holder** earning **RM3,000** is still flagged as **High Risk (74%)**. According to the 'Important Factors' graph, the long Red Bar indicates that while this salary attempts to signal for stability, it is ultimately overpowered by the Education factor (Green Bar).

This proves that for degree holders, earning RM3,000 is often insufficient to overcome the compounded financial burdens of student loans and the high cost of urban living.

Most Likely: High Risk



Important Factors for High Risk



[Click here to watch the demo video for the simulator.](#)



CONCLUSION & RECOMMENDATION

CONCLUSION

Malaysia's graduate talent crisis is driven by structural inequality rather than wages alone. The analysis shows that talent risk in Malaysia is highly asymmetric, concentrated especially among diploma holders face the largest expectation gap, where average salaries overstate what most actually earn, and rural graduates, who face a location-based salary penalty that fuels urban migration. At the same time, unequal opportunities persist, as Degree holders experience better employment outcomes while Diploma holders face limited progression and higher underemployment risk. These findings indicate that effective interventions require targeted strategies focused on these high-risk groups, rather than broad, uniform salary policies.

RECOMMENDATION

- **For Government:** Shift from blanket B40 aid to Targeted Upskilling Grants for high-risk urban graduates.
- **For Universities:** Use the model to audit and remove degrees that consistently lead to underemployment.
- **For Students:** A "Reality Check" tool to warn students if their expected salary doesn't match their urban cost of living.
- **For Employers:** A "Fair Wage" monitor to adjust salaries so employees stay out of the "High Risk" zone.

